

IIT Kharagpur

Department of Mathematics

Advanced Machine Learning Lab

(LTP- 0-0-3,CRD- 2)

Implementation of :

1. Breadth first search, Depth first search, Best first search, depth bounded search, A* search, uniform cost search [3 hours]
2. Decision trees, minimax, alpha-beta pruning [3 hours]
3. Clustering: k means, agglomerative hierarchical clustering, density-based spatial clustering of applications with noise [3 hours]
4. Genetic algorithms, Swarm algorithms: Particle Swarm Optimization, Ant Colony Optimization [3 hours]
5. Regression (Linear Model, Generalized Linear Model) [3 hours]
6. Gradient descent, Perceptron [3 hours]
7. Artificial Neural Networks and backpropagation [3 hours]
8. Convolutional Neural Networks [3 hours]
9. Recurrent Neural Networks [3 hours]
10. Long Short-Term Memory [3 hours]
11. Graph Neural Networks [3 hours]
12. Bandit Algorithms [3 hours]
13. Dynamic Programming methods in Reinforcement Learning [3 hours]
14. Monte Carlo tree search [3 hours]

References:

6. Pattern recognition and machine learning, Christopher M. Bishop, Springer, 2007.
7. Neural Networks and Deep Learning - A Textbook, Charu C. Aggarwal, Springer, 2018.
8. Deep Reinforcement Learning, Aske Plat, Springer, 2022.
9. Recurrent Neural Networks - From Simple to Gated Architectures, Fathi M. Salem, Springer, 2022.
10. Graph Neural Networks: Foundations, Frontiers, and Applications, Lingfei Wu, Peng Cui, Jian Pei, Liang Zhao (Eds.), Springer 2022.

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